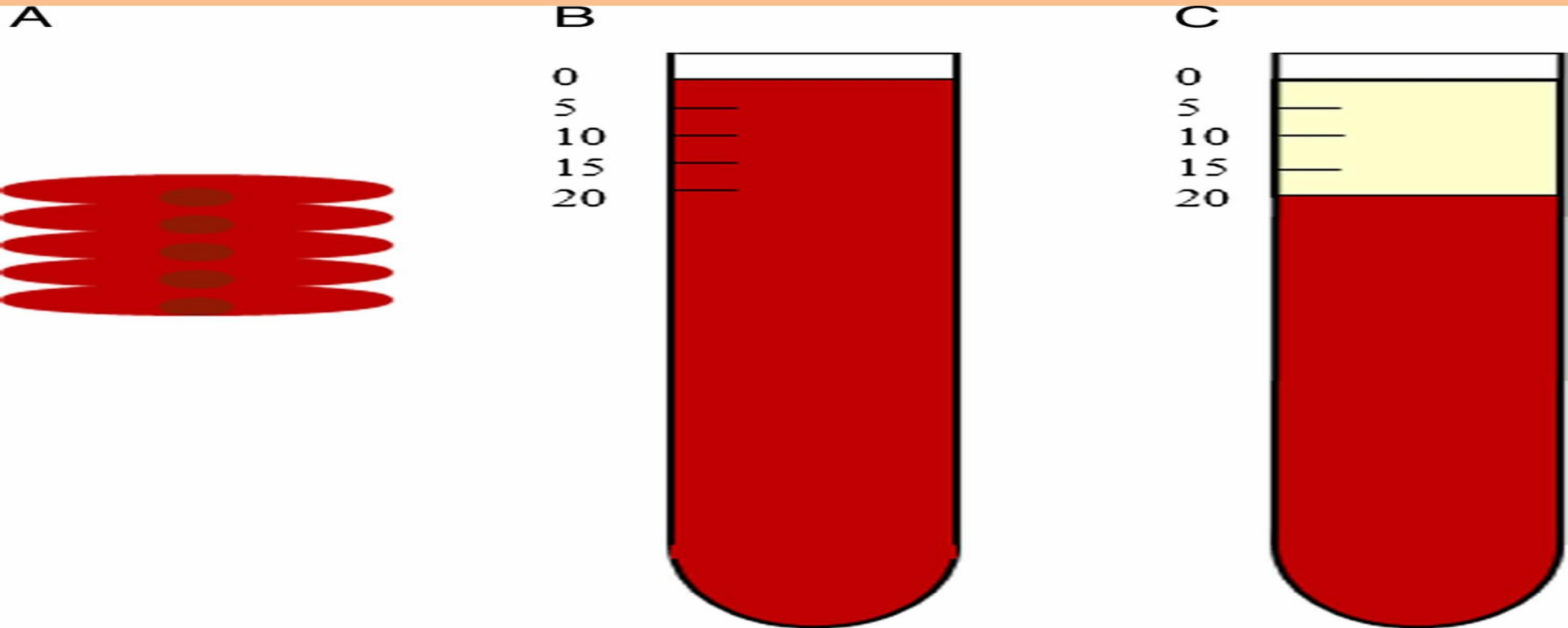


ERYTHROCYTE SEDIMENTATION RATE (ESR) & PACKED CELL VOLUME (PCV)



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LEARNING OBJECTIVES:

THE STUDENT SHOULD BE ABLE TO:

- ☐ Define the factors responsible for increasing or decreasing the ESR value
- ☐ Explain the importance of this practical in the prognosis of disease
- ☐ Read the values of ESR
- ☐ List the common causes of increased ESR

The Erythrocyte Sedimentation Rate (ESR)

It is a common hematological test for nonspecific detection of inflammation that may be caused by **infections**, some **cancers** and **certain autoimmune** diseases.

It can be defined as the,

Rate of sedimentation of red blood cells in a given interval of time is called erythrocyte sedimentation rate (ESR).

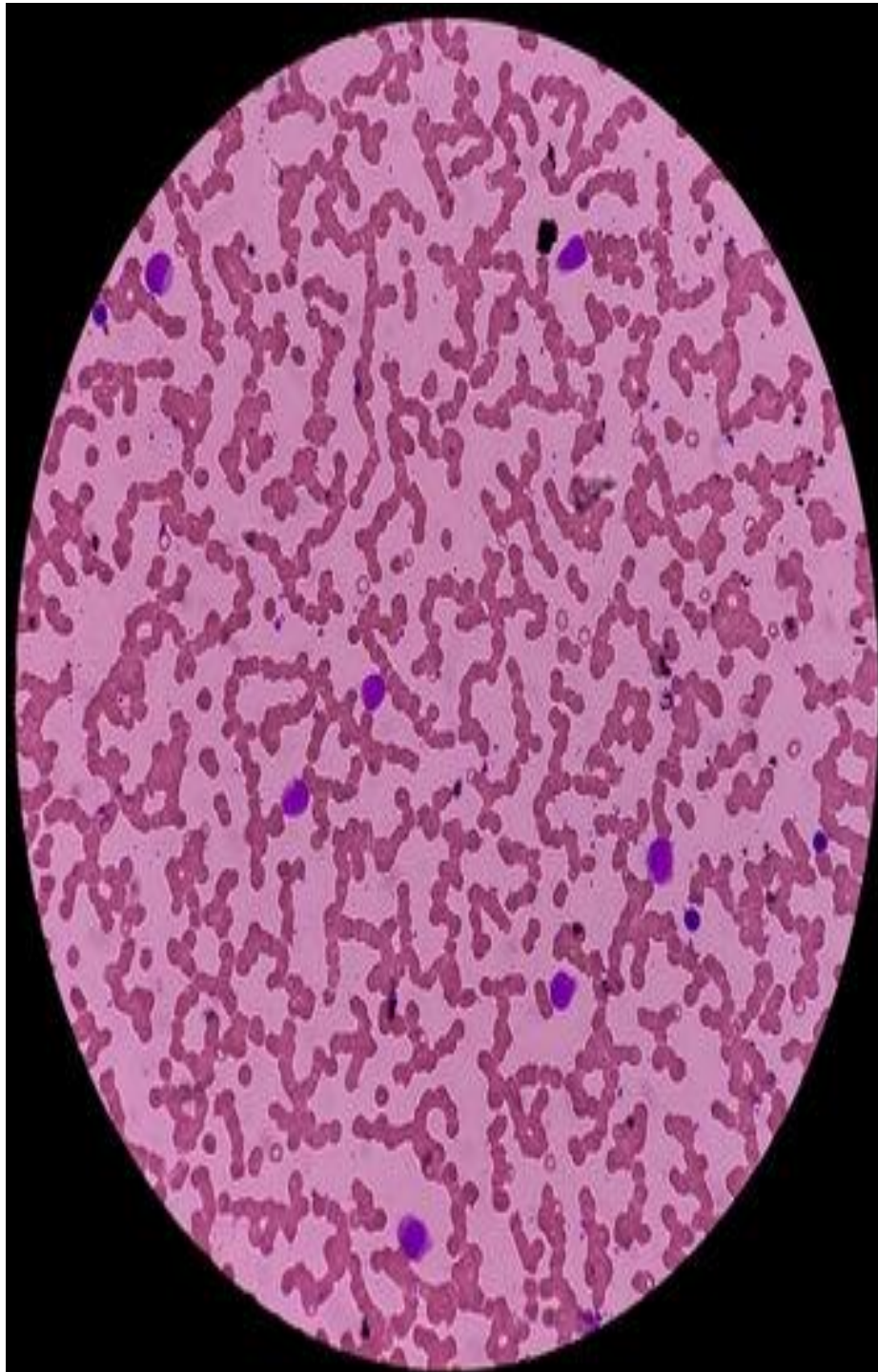
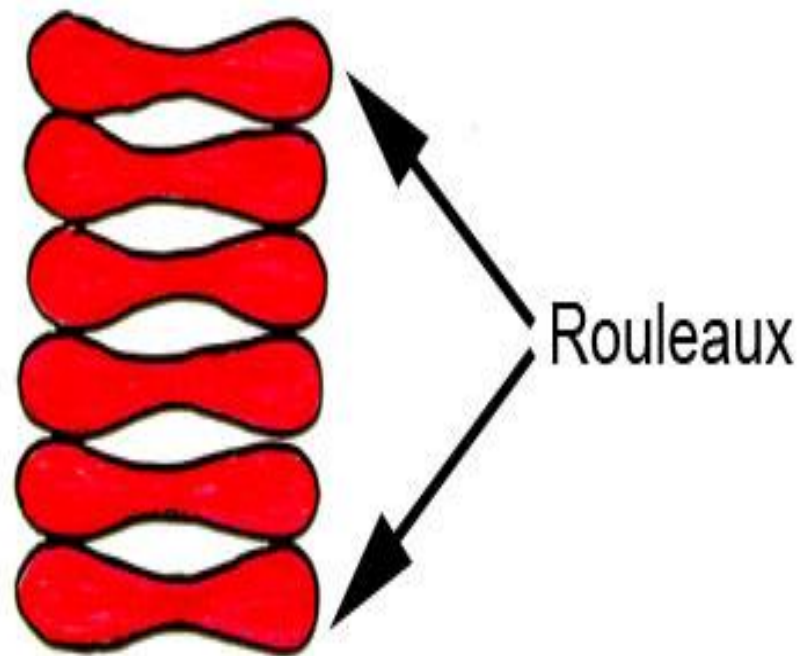
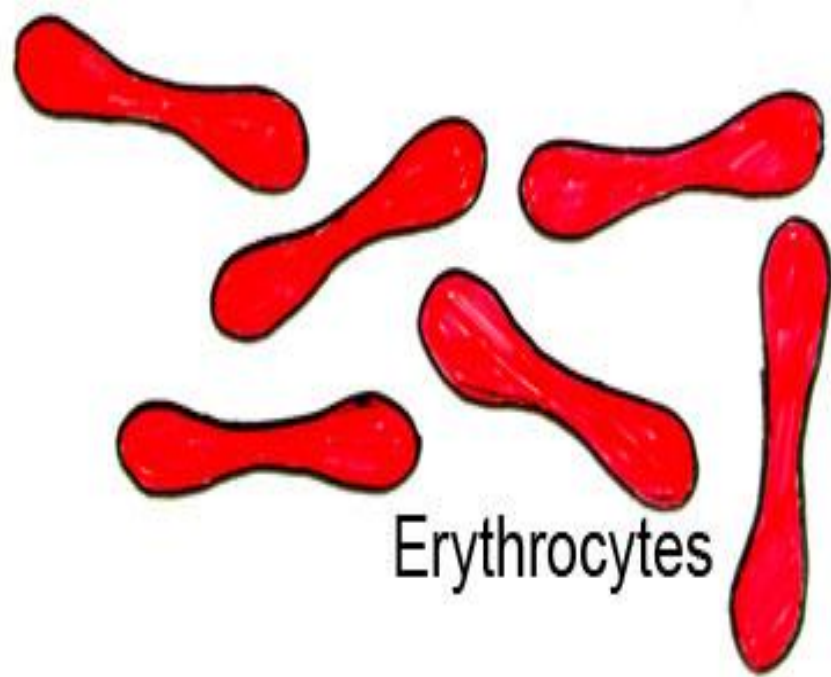
The normal given period of time is about one hour

WHAT IS THE ESR? (CONTINUED)

- ESR is the measurement of the suspension stability of red blood cells (RBCS) in plasma under specified test conditions. The two main determinants of ESR are
- degree of red cell aggregation and packed cell volume (hematocrit) ,the ratio of the volume of red blood cells to the total volume of blood .

PRINCIPLE OF EXPERIMENT:

- The principal of this experiment is to observe the rate of settling down of RBCs,
- due to **gravitational force** and
- **rouleaux formation**.



Principal of ESR:

When anticoagulated blood is allowed to stand in a narrow vertical glass tube, undisturbed for a period of time, the RBCs – under the influence of gravity- settle out from the plasma.

The rate at which they settle is measured as the number of millimeters of clear plasma present at the top of the column after one hour(mm/hr).

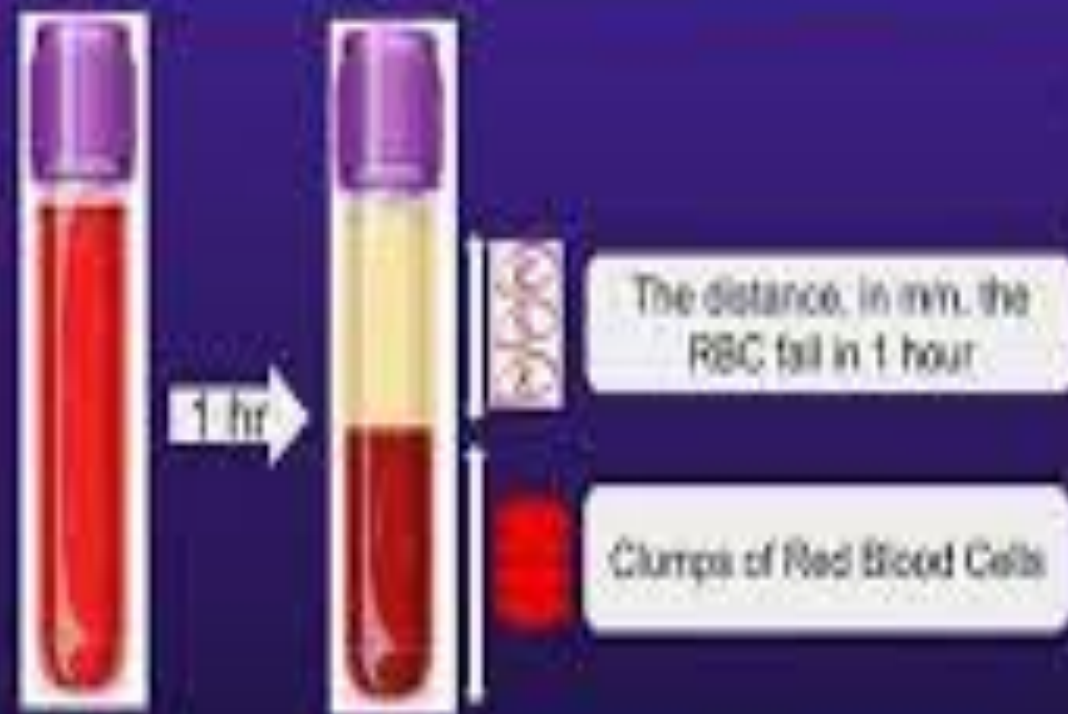
This mechanism involves three stages:

- **Stage of aggregation** : It is the initial stage in which piling up of RBCs takes place. The phenomenon is known as Rouleaux formation. It occurs in the first 10-15 minutes.
- **Stage of sedimentation** : It is the stage of actual falling of RBCs in which sedimentation occurs at constant rate. This occurs in 30-40 minutes out of 1 hour, depending upon the length of the tube used.
- **Stage of packing** : This is the final stage and is also known as stationary phase. In this, there is a slower rate of falling during which packing of sedimented RBCs in column occurs due to overcrowding. It occurs in final 10 minutes in 1 hour.

ESR measures the rate at which RBC settle in one hour at the bottom of a tube that contains a blood sample



The faster the blood cells sink, the more inflammation you have in your body



High ESR can be seen in:

- Inflammatory disease
- Infections
- Autoimmune diseases
- Chronic kidney disease
- Viral infections
- Pregnancy
- Cancer

Erythrocyte Sedimentation Rate Test

Factors affecting ESR

1-Size of RBC: in vitamin B12 deficiency RBC are large in size Macrocytes, and the sedimentation rate increased, While in iron deficiency the RBCs are small in size.

2-Number of RBCs:. In anemia; ESR increased while in polycythemia; ESR. decreased,

3-Plasma protein concentration: The more plasma protein present, the more rapidly rouleaux phenomena will occur and the more elevated the sedimentation rate will be.

4-Hemoglobin content .

5-blood viscosity

6-Rouleaux formation A rouleaux formation will give a false value

Clinical Applications

ESER Increased in:

1. chronic condition such as Rheumatoid arthritis and Tuberculosis .
2. A cancer of a certain type of white cells Multiple myeloma, where the plasma proteins are abnormal.
3. Acute infections .
4. Anemia.
5. kidney disease..
6. Autoimmune and inflammatory diseases .
7. Pregnancy .

Clinical Applications

ESR Decreased in:

- Any disease that changes the shape and size of red blood cells decrease the ESR;
- also the diseases that cause the body to make less protein or extra RBCs decrease the ESR, such as:

1. Sick cell disease..

2. Polycythemia

3. Congestive heart failure .

Clinical Significance of ESR

Because the ESR is changed in a great variety of conditions ,

Its alteration is not specific and not diagnostic.

It is a prognostic test :

1-It detects the presence and severity of disease.

2-It gives an idea about the activity of disease

3-Repeated ESR estimation helps in prognosis and follow up of disease

FACTORS AFFECTING THE ESR,

SUMMARIZED:

Increased ESR	Lower ESR	No effect
Old age	Extreme leucocytosis	Obesity
Female	Polycythaemia	Body temperature
Pregnancy	Spherocytosis, microcytosis	Recent meal
Anaemia	Hyperviscosity	Aspirin, NSAIDs
Paraprotein (myeloma)	Low protein: fibrinogen, gammaglobulins	
Hypergammaglobulinaemia		
Macrocytosis	Technical factors: dilution, clotted sample	
Elevated fibrinogen (infection, inflammation, malignancy)		
Technical factors: dilution, high temperature		

ESR TEST RESULTS

NORMAL RANGE



MEN

0-15 MM / HR



WOMEN

0-20 MM / HR



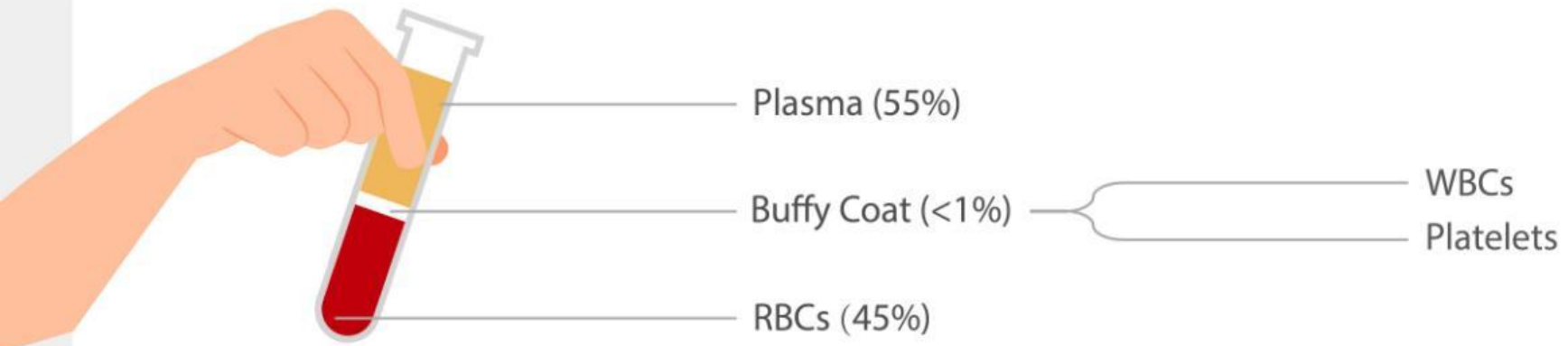
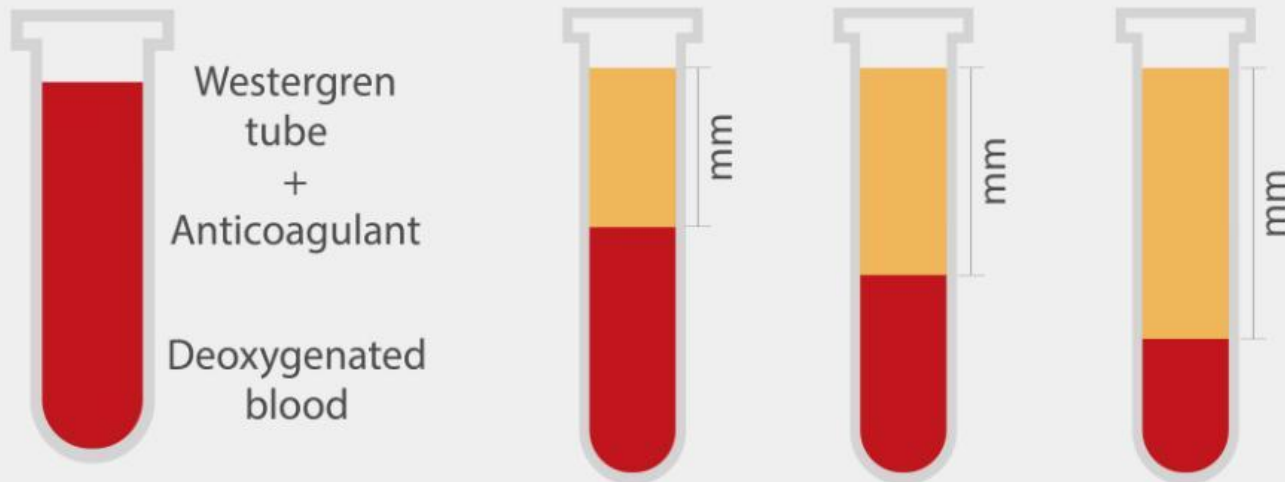
CHILDREN

0-20 MM / HR





When blood is left alone in an upright test tube, the sedimentation increases over time, but the ESR stays the same, which indicates how quickly the sedimentation happens.



THE WINTROBE SEDIMENTATION RACK:



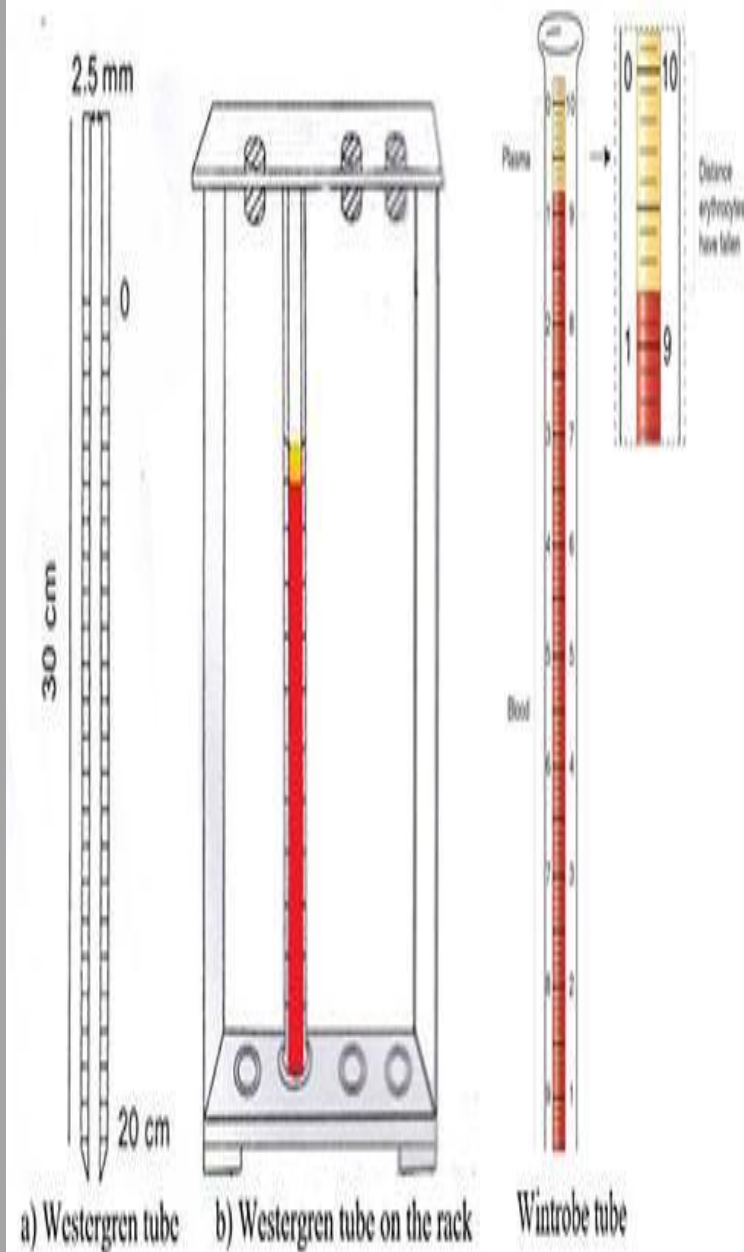
HOW IS IT CARRIED OUT?

- When anticoagulated whole blood is allowed to stand in a narrow vertical tube for a period of time, the RBCs – under the influence of gravity - settle out from the plasma.
- The rate at which they settle is measured as the number of millimeters of clear plasma present at the top of the column after one hour (mm/hr).



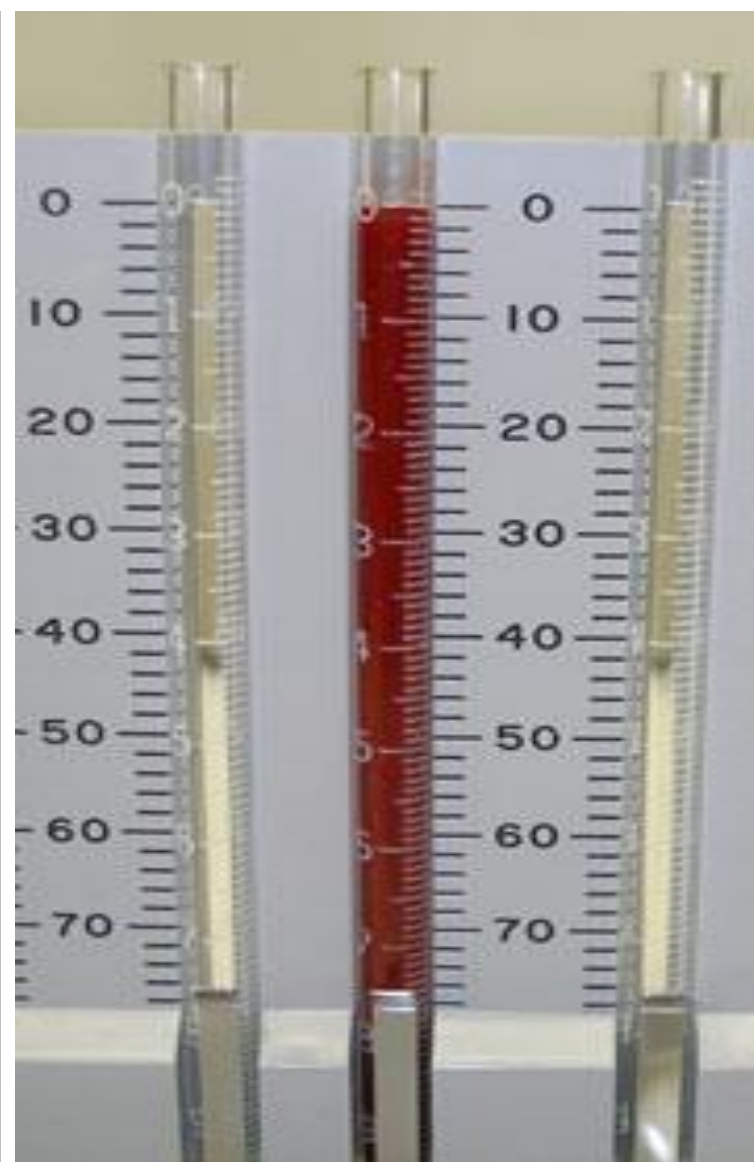
WESTERGREN METHOD:

- The Westergren method measures the distance (in millimeters) at which red blood cells in anticoagulated whole blood fall to the bottom of a standardized, upright, elongated tube over one hour due to the influence of gravity.
- The tube used for the test is called the Westergren tube.
- The tube is placed in a rack in a strictly vertical position for 1 hour at room temperature, at which time **the distance from the lowest point of the surface meniscus to the upper limit of the red cell sediment** is measured.
- The distance of fall of erythrocytes, expressed as millimeters in 1 hour, is the ESR.



WINTROBE METHOD:

- The Wintrobe method is performed similarly except that the Wintrobe tube is **smaller in diameter** than the Westergren tube and **only 100 mm long**.
- EDTA anticoagulated blood without extra diluent is drawn into the tube, and the rate of fall of red blood cells is measured in millimeters after 1 hour.
- The shorter column makes this method **less sensitive** than the Westergren method because the **maximal possible abnormal value is lower**.



This picture shows a rack holding Wintrobe tubes, in which anticoagulated whole blood has just been added. (Time: 0)

ACADEMIC IMPLICATIONS:

Why RBCs sediment

The RBCs sediment because their density is greater than that of plasma; this is particularly so, when there is an alteration in the distribution of charges on the surface of the RBCs (which normally keeps them separate from each other) resulting in their coming together to form large aggregates known as rouleaux.

PROCEDURE:

- This experiment is performed by Westergren method.
 - The Westergren sedimentation tube is 300mm in length and diameter of about 2.5 mm
 - The lower $2/3^{\text{rd}}$ is graduated from 0-200mm and upper $1/3^{\text{rd}}$ is left unmarked. It is open at both ends.
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- ❖ Obtain 2 ml of blood by venepuncture.
 - ❖ Take 0.4ml of 3.8% of Na-Citrate (anticoagulant) and add 1.6ml of blood and mix it
 - ❖ Suck this blood in Westergren tube upto 0 mark very carefully by suction device.
 - ❖ Put the tube in the ESR stand and leave it undisturbed.
 - ❖ Read the column of clear plasma after 1^{st} hour.

Q-1-Define erythrocyte sedimentation rate?

The Erythrocyte Sedimentation Rate (ESR) is a blood test that measures how quickly red blood cells settle to the bottom of a test tube, indicating the presence and degree of inflammation in the body. It's a nonspecific indicator, meaning it can be elevated due to various conditions, but is often used to monitor inflammatory diseases and infections.

A faster settling rate, or higher ESR, often indicates inflammation or infection in the body. This happens because proteins associated with inflammation cause red blood cells to clump together and settle more rapidly.

Q-2-What is the cause of sedimentation?

Sedimentation, the process where particles settle out of a fluid, is primarily caused by gravity acting on particles suspended in a fluid like water or air.

Q-3-How does it increase in disease?

In disease, sedimentation, specifically the erythrocyte sedimentation rate (ESR), increases due to the presence of inflammatory proteins in the blood. These proteins cause red blood cells (RBCs) to clump together, forming Rouleau, which settle faster than individual RBCs, thus increasing the ESR.

Q-4-What is the clinical importance of ESR?

An erythrocyte sedimentation rate (ESR) is a blood test that can show if you have inflammation in your body. Inflammation is your immune system's response to injury, infection, and many other conditions, including immune system disorders, certain cancers, and blood disorders

Q-5-Why is the ESR value higher in females as compared to males?

ESR (Erythrocyte Sedimentation Rate) is typically higher in females than in males due to a combination of physiological and hormonal factors. These factors include differences in red blood cell counts, hormonal fluctuations related to menstruation, pregnancy, and menopause, and the higher prevalence of certain autoimmune diseases in women.

THANK YOU

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